Hypertension

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Definition

Classification of hypertension is presented in **Table 1**.

TABLE 1: Classification of hypertension.

Classification	*Children aged 1–12 years (percentile based)*	*Adolescents ≥13 years (mm Hg based)*
Normotensive	<90th percentile	<120/<80 mm Hg
Elevated BP (previously called prehypertension)	≥90th percentile to <95th percentile or 120/80 mm Hg to <95th percentile (whichever is lower)	120/<80 to 129/<80 mm Hg
Stage 1 hypertension	≥90th percentile to <95th percentile +12 mm Hg or 130/80 to 139/89 mm Hg (whichever is lower)	130/80 to 139/89 mm Hg
Stage 2 hypertension	≥95th percentile + 12 mm Hg, or ≥140/90 mm Hg (whichever is lower)	≥95th percentile + 12 mm Hg, or ≥140/90 mm Hg (whichever is lower)

(BP: blood pressure)

Incidence

- ☑ Prevalence is 7% for hypertension (HTN), 4% for sustained HTN and 10% for elevated HTN in a recent study of the Indian school children. However, as per the data from large studies, the overall prevalence of HTN in children and adolescents is around 3.5% with a higher prevalence in overweight and obese.
- ☑ The incidence of pediatric primary HTN is increasing and it should be identified early in primary care settings. If left untreated, HTN in children and adolescents can have significant implications for cardiovascular and renal health into adulthood, including metabolic syndrome, stroke, coronary artery disease, chronic kidney disease, and heart failure. Both elevated blood pressure (BP) and HTN affect boys more than girls, with rates among adolescents higher than those for younger children.

Confirm Hypertension

- ☑ In a patient with abnormal BP, measurements should be taken twice during the same visit. While oscillometric measurements are good for screening, hypertension should always be confirmed with auscultatory method. A diagnosis of hypertension should be made with three separate BP recordings at least a week apart (hypertensive emergency being an exception). If the elevation persists at repeat measurement, recommend lifestyle interventions and repeat BP measurements at 6 and 12 months. If BP remains elevated after 12 months, order "ambulatory BP monitoring (ABPM)". Hypertension should only then be diagnosed if the patient's BP is elevated at three separate visits.
- ☑ Increased stress on importance of ABPM in diagnosis and management of childhood hypertension.
- ☑ ABPM has been strongly recommended for confirming a diagnosis of HTN in children and adolescents if they have office BP measurements in the elevated BP category for 1 year or more or with stage 1 HTN over three clinic visits.
- ☑ ABPM should be done for suspected white-coat HTN or masked HTN.
- ☑ Its use was particularly recommended in special group of populations such as chronic kidney disease and post transplantation.

Frequency of Monitoring

- ☑ The guidelines recommend annual BP measurement for children above 3 years and has identified sub-groups (obese, on medications known to increase BP, especially steroids, renal disease, history of coarctation, or diabetes) for more frequent checks.
- ☑ Children younger than 3 years warrant regular measurements if they have any of the following: congenital heart disease, recurrent urinary tract infection, urological malformation, solid-organ transplant, bone marrow transplant, malignancy, neurofibromatosis, tuberous sclerosis, or sickle cell disease. Small for gestational age newborns, premature (<32 weeks), or very low birth weight babies and those with umbilical arterial catheterization also require regular checks.

Elevated Hypertension

Prehypertension is now labeled as elevated BP.

Primary Hypertension

- ☑ Primary or essential HTN is the predominant form of HTN in 6–12-year-olds and adolescents, especially in those with a family history of HTN or who are overweight and/or obese.
- ☑ Children with primary HTN are mostly overweight and with a family history positive for HTN. In obese patients, screening should include hemoglobin A1C, lipid panel, serum creatinine, and liver enzymes.
- ☑ Children age 6 years and older do not require an extensive evaluation for secondary causes of HTN if they have a positive family history of HTN, are overweight or obese, and/or do not have history or physical examination findings suggestive of a secondary cause of HTN. Echocardiography is recommended to assess for cardiac target organ damage, especially left ventricular hypertrophy at the time that pharmacologic treatment of HTN is considered in any child.

Secondary Hypertension

- ☑ Secondary HTN is more common in pediatric age group (especially children <6 years of age) except for adolescents and obese.
- ☑ Secondary HTN is primarily caused by renal and/or renovascular disorders, which account for 63–74% of the cases.
- ☑ Factors causing secondary HTN in children commonly are renal parenchymal disease, renal artery stenosis, coarctation of the aorta, environmental exposures, neurofibromatosis, hormonal causes, sleep apnea, and medications.
- ☑ Based on patient history and physical examination findings, a urinalysis, urine chemistry panel, thyroid function tests, sleep study, complete blood cell (CBC) count, and/or a drug screen may be warranted. An elevated diastolic BP is more predictive of secondary HTN (specifically renovascular disease), which requires renal ultrasound. Subsequent laboratory abnormalities may prompt further workup of renal, cardiac, or endocrine disorders as secondary causes of HTN.

Hypertensive Crisis

Hypertensive crisis is defined as a severe elevation in BP, classified as hypertensive emergencies and urgency.

Hypertensive urgency, in which there are no signs of end-organ damage, and *hypertensive emergency*, in which signs of end-organ damage are present, such as distinct signs of hypertensive encephalopathy, acute left ventricular failure and acute myocardial ischemia, papilledema, and elevated liver function tests.

Echocardiography

Echocardiography is strongly recommended to assess for target organ damage at the time of consideration of pharmacologic treatment of HTN.

Treatment

Lifestyle

- ☑ *Diet:* Dietary approaches to stop hypertension (DASH) diet that includes increased fruits, vegetables, and grains and low in sodium and fats
- ☑ *Exercise:* 40 minutes of moderate aerobic activity 3 days per week
- ☑ Motivational interviewing and counseling pharmacologic.

Target BP: Reduction in systolic and diastolic BP to <90th percentile.

Stepwise Approach to Hypertension

Normotensive

No additional action needed. Measure the BP at the next routine Well-child Care Visit.

Elevated Blood Pressure (Previously called Prehypertension)

- ☑ *Step 1:* Lifestyle modification; repeat BP after 6 months.
- ☑ *Step 2:* If still elevated, check all four-limb BP [upper limb/lower limb (UL/LL)]. If these are normal—lifestyle modification continued and BP rechecked again after 6 months.
- ☑ *Step 3:* If BP still elevated, ABPM should be ordered (if available), and consider diagnostic evaluation. If BP normalizes at any point, return to annual BP screening at Well-child Care Visits.

Stage 1 Hypertension

- ☑ *Step 1:* If asymptomatic, provide lifestyle counseling and recheck the BP in 1–2 weeks by auscultation.
- ☑ *Step 2:* If the BP reading is still at the stage 1 level, UL/LL BP should be checked and if normal nutrition and/or weight management initiated and BP rechecked in months.
- ☑ *Step 3:* If BP continues to be at stage 1 HTN level after three visits, ABPM should be ordered (if available), diagnostic evaluation should be conducted, and treatment should be initiated. Subspecialty referral should be considered. If symptomatic, early initiation/referral should be considered.

Stage 2 Hypertension

Patients with stage 2 HTN or with target organ damage should be started on antihypertensives immediately and all efforts should be made to identify an underlying cause and the patient should be referred to subspecialty care within a week.

Choice of Medications

First-line Agent

If patient has chronic kidney disease, proteinuria, or diabetes, angiotensin-converting enzyme (ACE) inhibitors (captopril, enalapril, fosinopril, lisinopril, ramipril, or quinapril) or angiotensin II receptor blockers (candesartan, irbesartan, losartan, olmesartan, and valsartan) or long-acting calcium channel blockers (amlodipine, felodipine, and isradipine) are recommended. Nifedipine is used only for short-term control of BP.

Second-line Agent

- ☑ If second-line agent needed (also can be first-line): thiazide diuretic (chlorthalidone, chlorothiazide, and hydrochlorothiazide).
- ☑ If the patient has heart failure, if first-line treatments fail to work, or if the patient has a specific diagnosis for which beta-blockers are indicated, prescribe beta-blockers (atenolol and metoprolol). If the HTN is not controlled on maximum doses of these drugs consider peripheral alpha receptor blocker like prazosin or central adrenergic agonist like clonidine.

Management of Hypertensive Crisis

- ☑ Pediatric patients with hypertensive crisis require immediate and appropriate reduction in BP levels, whereas patients with hypertensive urgency require a slower rate of reduction in BP levels over 24–48 hours. However, rapidly decreasing BP levels may result in decreasing the blood flow of organs, causing ischemia and infarction.
- ☑ The treatment goal with pharmacologic therapy should be a reduction in systolic blood pressure (SBP) and diastolic blood pressure (DBP) to <90th percentile and <130/80 mm Hg in adolescents ≥13 years of age. The rate of BP reduction should be 25% over a period of 6–8 hours, which is gradually reduced to normal over 24–72 hours since sudden, drastic reductions in BP can itself contribute to organ damage secondary to ischemia.
- ☑ In patients with hypertensive encephalopathy combined with chronic HTN, it is important to reduce the mean arterial pressure gradually during the first hour. The preferred agents include infusion of labetalol, esmolol, and nicardipine. Intravenous (IV) sodium nitroprusside infusion can also be used for hypertensive emergency **(Table 2)**.

TABLE 2: Medicines for hypertensive crisis.

Drug	*Class*	*Onset (minutes)*	*Duration*	*Route*	*Dose*
Labetalol	α + β blocker	2–5	2–12	IV bolus	0.2–1 mg/kg/dose, up to 40 mg/dose; infusion 0.25–3 mg/kg/h
Nicardipine	Calcium channel	10	<8	IV bolus or infusion	Bolus: 30 µg/kg up to 2 mg/dose; infusion: 0.5–4 µg/kg/min
Hydralazine	Arterial vasodilator	5–20	1–4	IV/IM bolus	IV: 0.2–0.6 mg/kg
Esmolol	Beta-1 blocker	2–10	10–30 (minutes)	IV infusion	100–500 µg/kg/min, up
Sodium nitroprusside	Vasodilator	2–10	1–10 (minutes)	IV infusion	0.5–10 µg/kg/min
Fenoldopam	Dopamine receptor agonist	10	1	IV infusion	0.2–0.8 µg/kg/min
Enalapril	ACE inhibitor	15–30	6–12	IV bolus	5–10 µg/kg/dose up to

(ACE: angiotensin-converting enzyme; IM: intramuscular; IV: intravenous)

Referral

Nephrology or cardiology, pending laboratory results and after a diagnosis of HTN has been confirmed.

Drug classes that can be used in children with hypertension are presented in **Table 3**.

TABLE 3: Drug classes used in children with hypertension.

Class	*Drug*	*Initial dose*	*Dose per 24 hours*	*No. dose/ 24 hours*
Calcium channel blockers	Amlodipine	0.1–0.2 mg/kg Maximum 5 mg	0.6 mg/kg/d Maximum 10 mg/d	1
	Nicardipine LP	0.25–0.5 mg/kg/d	1–3 mg/kg/d Maximum 120 mg/d	2
	Felodipine	5 mg	10 mg	1

Contd...

Referral

Contd...

Class	*Drug*	*Initial dose*	*Dose per 24 hours*	*No. dose/ 24 hours*
ACE inhibitors	Captopril	☑ *Liquid formulation:* 5 mg/ 5 mL and 25 mg/5 mL ☑ *Newborns:* 0.01–0.03 mg/kg/dose ☑ *Infants:* 0.1 mg/kg/dose ☑ *Older children:* 0.3–0.5 mg/kg/dose	6 mg/kg/d Maximum 150 mg	1 to 3
	Enalapril	☑ *Newborns:* 0.05 mg/kg/dose ☑ *Infants:* 0.05 mg/kg/dose ☑ *Older children:* 0.08 mg/kg/dose ☑ Maximum 5 mg	0.8 mg/kg/d Maximum 40 mg/d	1
	Lisinopril	0.08 mg/kg Maximum 5 mg/d	0.6 mg/kg/d Maximum 40 mg/d	1
ARBs	Losartan*	0.7 mg/kg/d Maximum 50 mg/d	1.4 mg/kg/d Maximum 100 mg/d	1
	Irbesartan	2 mg/kg/d	6–12 years <35 kg: 75–150 mg/d ≥13 years >35 kg: 150–300 mg/d	1
	Valsartan	3 mg/mL drug with liquid formulation	1–6 years: 1 mg/kg/d and <4 mg/kg/d 6–18 years: <35 kg: 20 mg/d and <40 mg/d >35 kg: 40 mg/d and <80 mg/d	
Beta-blockers	Acebutolol	1.5–3 mg/kg/d	5–15 mg/kg/d	1 to 2
	Propranolol	1 mg/kg/d	4 mg/kg/d Maximum 640 mg/d	2 to 3
	Atenolol	0.1–1 mg/kg/d	2 mg/kg/d Maximum 100 mg/d	1 to 2
Alpha- and beta-blocker	Labetalol	1–3 mg/kg/d	10–15 mg/kg/d Maximum 1,200 mg/d	2
Alpha-blockers	Prazosin	0.05–0.1 mg/kg/d Maximum 0.5 mg × 2 per day	0.5 mg/kg/d Maximum 20 mg/d	2 to 3
	Clonidine	5 µg/kg/d	30 µg/kg/d Maximum 1.05 mg/d	2 to 3
Diuretics	Hydrochloro-thiazide	0.5–1 mg/kg/d	3 mg/kg/d Maximum 50 mg/d	1
	Furosemide Spirono-lactone*	0.5–2 mg/kg/dose 1 mg/kg/d	6 mg/kg/d 3.3 mg/kg/d Maximum 100 mg/d	1 to 2 1 to 2

(ACE: angiotensin-converting enzyme; ARBs: angiotensin receptor blockers)

Further Reading

- ☑ Flynn JT, Kaelber DC, Baker-Smith CM, Blowey D, Carroll AE, Daniels SR, et al. Clinical practice guideline for screening and management of high blood pressure in children and adolescents. Pediatrics. 2017;140(3):e20171904.
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